US Inflation Forecast, Oct 2026 – Sep 2027

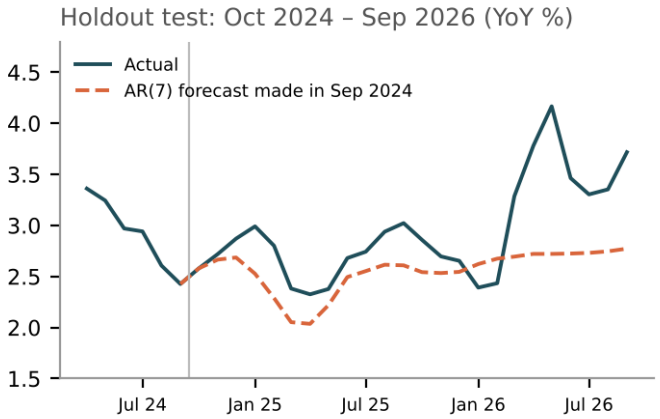
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Question & method

Where is US consumer price inflation headed over the next 12 months? I used 153 months of official CPI data, tested the series for stationarity, chose an autoregressive (AR) time-series model, checked it on two years of data it had not seen, and then forecast to September 2027. October 2025 has no CPI reading because BLS did not publish one during the federal government shutdown; I filled that month with the geometric mean of September and November (324.654).

CPI inflation, Sep 2026	Forecast peak	Forecast, Sep 2027	95% range, Sep 2027
3.7%	**4.7% (Jul 2027)**	**4.2%**	**0.7% – 7.7%**





Results

Step	Result	What it means
Stationarity (ADF test)	Log CPI: τ = -1.50 (5% critical -3.44). Monthly inflation: τ = -6.48 (5% critical -2.88).	Prices trend upward without settling, but monthly inflation is stable, so the model is built on inflation.
Model choice	Schwarz picks AR(1), but its residuals fail the Ljung-Box test (p = .005). Akaike picks AR(7), which passes (p = .245).	AR(7) captures patterns AR(1) leaves behind. It is stable (largest root 0.93 < 1).
Final AR(7) (N = 145)	R^2 = 0.39; strongest terms: last month (b = 0.58, p < .001) and 7 months back (b = 0.29, p = .001). Implied long-run rate 3.2% a year.	Inflation has momentum: a hot month tends to be followed by another.
Holdout accuracy (monthly RMSE)	AR(7) 0.248 vs last-12-month average 0.253 vs 2%-target pace 0.265 (percentage points).	The model was 6% more accurate than assuming 2% and 2% more accurate than recent trend: a modest gain.

What this means

- **Inflation is likely to stay well above the Fed's 2% target.** The forecast holds year-over-year inflation between 3.6% and 4.7% through September 2027, up from 3.7% in September 2026, because the strong price increases of spring 2026 carry forward.
- **The uncertainty is wide.** By September 2027 the 95% range runs from 0.7% to 7.7%, so the forecast is best read as a central path, not a promise.
- **The holdout test shows the main risk:** a model trained to September 2024 expected about 2.8% inflation by September 2026, but the actual figure was 3.7%. Univariate models miss shocks such as the 2026 price jump.

Business use

For a retailer such as a pharmacy, plan for supplier costs to rise about 4% over the next year, review shelf prices quarterly rather than annually, and update this forecast each month when BLS releases new CPI data.

Limitations: an AR model uses only past inflation, not interest rates, oil prices or wages. Residuals are not normally distributed (Jarque-Bera p < .001, driven by the 2020–22 shocks), so the intervals are approximate. October 2025 is interpolated, which slightly affects the October 2026 year-over-year figure.